

GridCast: Probabilistic Grid Stress Forecasting for Dynamic Data Center Power Allocation

Short Technical Report

Nabil Zouhari Rayane Boussanni

Demo: <https://gridcast.dev/> | Walkthrough: <https://www.youtube.com/watch?v=-OHXRK749SA>

Abstract

About 40% of the U.S. electricity grid’s installed capacity sits idle on a typical day, held back as insurance against the few hundred hours per year when weather, demand and forecast error collide. Co-locating AI data centers next to that idle capacity is now the dominant new load on the grid, but pricing flexible power contracts between utilities and AI tenants requires multi-day demand forecasts with calibrated uncertainty, a regime where the U.S. Energy Information Administration’s (EIA) official day-ahead forecast holds only 7.57% MAPE with no probabilistic envelope. We present GRIDCAST, a Temporal Fusion Transformer (TFT) trained jointly across four major U.S. transmission nodes that produces 10-day quantile forecasts of hourly demand from weather and calendar inputs, and translates those forecasts into a single operator-facing power-allocation percentage. On a 60-day rolling holdout the model achieves 4.50% MAPE, 41% lower than the EIA day-ahead baseline, and holds 4.65% MAPE at days 8 to 10 (better than 24-hour persistence at day 1). Post-hoc per-node calibration restores nominal p_{10} to p_{90} coverage from 58.1% to 80.0%.

1 Introduction and Problem

The U.S. grid is overbuilt by design. EIA load-factor data for 2025 implies an average national load of 59.8% of peak, so roughly 40% of installed capacity sits unused on an average day. That excess is not waste in the colloquial sense: it is operational reserve against the rare hours when weather and demand together push the system to its edge. But it is also a vast pool of otherwise-saleable capacity that no utility can confidently bring to market, because the same uncertainty that justifies the reserve also prevents underwriting any contract against it.

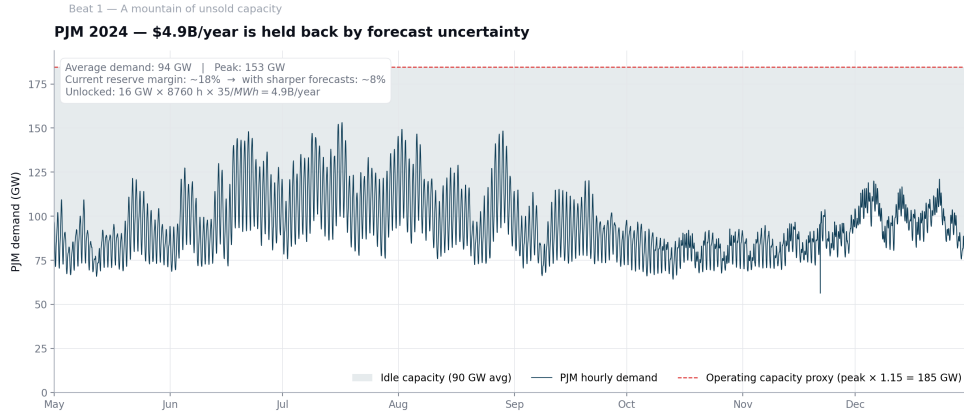


Figure 1: Hourly demand in PJM through 2024 against an operating capacity proxy. The shaded band averages ~90 GW of idle capacity. The order-of-magnitude estimate inside the figure (tightening reserves from ~18% to ~8% by sharpening the demand forecast unlocks roughly $16 \text{ GW} \times 8760 \text{ h} \times \$35/\text{MWh} \approx \$4.9\text{B}$ per year in PJM alone) sets the economic stake.

The contested side of this idle capacity is the rapid build-out of AI data centers. A single AI training campus now draws hundreds of megawatts, and multi-gigawatt power purchase agreements at hyperscaler scale are being signed annually. AI workloads are unusually well-suited to flex with the grid (training is interruptible, batch inference can be deferred, inference traffic can be shifted geographically), yet they cannot do so without a forecast that an operator can underwrite. The product we describe is the forecasting layer that makes such a contract pricable.

Three requirements follow directly. The forecast must be (i) *probabilistic*, since stress emerges from compounding tail risk and a point estimate cannot price an option, (ii) *calibrated*, so that “80% confidence” actually means 80%, and (iii) *accurate over a multi-day horizon*, since a 24-hour outlook does not give a data center time to drain a training queue or reroute inference.

2 Why Weather, and Why a TFT

Weather is the dominant signal. Demand at each of the four nodes we model is a smooth, non-linear function of temperature. Figure 2 shows the relationship on four years of hourly data: each region exhibits the classical V or U-curve in which both extreme cold and extreme heat raise load. The dispersion around the curve is the irreducible component that a probabilistic forecaster has to quantify.

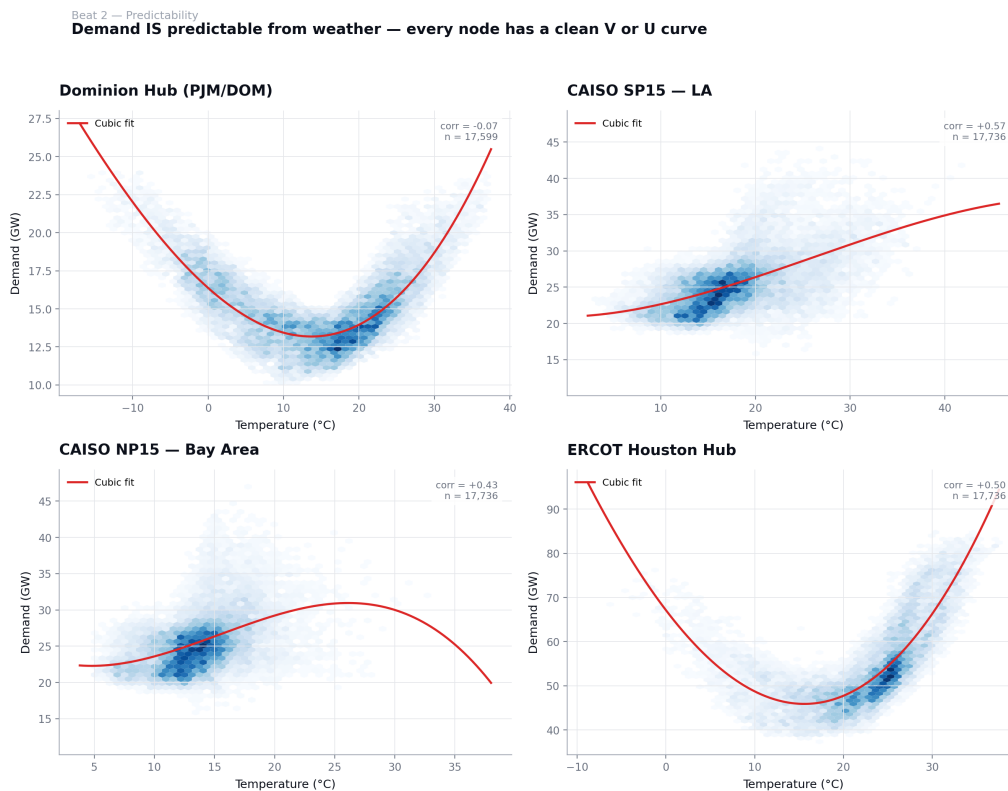


Figure 2: Hourly demand against 2 m temperature at each modeled node, $n \approx 17,700$ hours per panel. Red curves are cubic fits. The clean V/U shape is what makes weather-driven forecasting tractable; the residual scatter is the irreducible noise that probabilistic forecasting must absorb.

Why a Temporal Fusion Transformer. The Temporal Fusion Transformer (TFT) [1] is a sequence-to-sequence architecture designed for multi-horizon probabilistic forecasting with heterogeneous inputs. We chose it over plain LSTM, vanilla transformer, and gradient-boosted alternatives for three reasons.

1. **Heterogeneous input typing.** The TFT explicitly distinguishes *static* covariates (the node identity), *time-varying known-future* covariates (weather forecasts and calendar features) and *time-varying unknown* covariates (past demand only). A vanilla LSTM cannot natively condition future predictions on *future*

weather, which is the strongest signal we have for the 10-day outlook.

2. **Native quantile output.** Trained with the pinball loss across $\mathcal{Q} = \{0.10, 0.25, 0.50, 0.75, 0.90\}$, the model produces a probabilistic envelope rather than a point estimate. This is the signal the downstream allocation policy depends on.
3. **Multi-horizon in one pass.** A single forward pass produces the entire 240-hour curve, which avoids the error compounding of autoregressive prediction over a 10-day window.

3 Method

Setup. A single joint TFT is trained across the four nodes {Dominion Hub, CAISO SP15, CAISO NP15, ERCOT Houston} with `node_id` as a static categorical. Encoder length is 168 hours (7 days of history); decoder length is 240 hours (10 days). Inputs are hourly EIA Balancing-Authority demand and Open-Meteo weather (2 m temperature, dewpoint, humidity, wind, cloud cover, radiation, precipitation) plus calendar features. Hyperparameters: hidden size 32, attention heads 4, dropout 0.1, Adam at 3×10^{-3} , batch size 64, gradient clipping 0.1, early stopping on validation loss.

Quantile loss. The TFT is trained to minimize the pinball loss, summed over quantile levels $q \in \mathcal{Q}$, horizon steps $h \in \{1, \dots, 240\}$, and training samples:

$$\mathcal{L}_q(y, \hat{y}_q) = \max(q(y - \hat{y}_q), (q - 1)(y - \hat{y}_q)), \quad \mathcal{L}_{\text{total}} = \sum_{q \in \mathcal{Q}} \sum_{h=1}^{240} \mathcal{L}_q(y_h, \hat{y}_{q,h}). \quad (1)$$

The asymmetric penalty makes $\hat{y}_q$ converge to the conditional q -th quantile of the target, which is what gives the model a calibrated forecast envelope rather than just a mean.

Post-hoc calibration. Even with the pinball loss, the raw model is over-confident: the nominal p_{10} to p_{90} band covers only 58.1% of held-out actuals. We apply a per-(node, horizon-bucket) additive correction learned on a calibration window, which lifts empirical coverage to the nominal 80%.

From forecast to allocation. At each forecast issue time t , we compute the model-estimated probability that future hourly demand exceeds a node-specific stress threshold, evaluated at the calibrated p_{90} quantile, and aggregate that across the 240-hour decoder window. Call this scalar $s_{p_{90}}(t) \in [0, 1]$. The product output is the single percentage

$$\text{ALLOCATION}(t) = \text{round}\left((1 - s_{p_{90}}(t)) \cdot 100\right) \in \{0, 1, \dots, 100\}. \quad (2)$$

The interpretation is operational: “with 90% confidence, the node can safely commit ALLOCATION% of its excess capacity to flexible tenants over the forecast horizon.”

4 Results

Evaluation protocol. A 60-day window at the end of available data is held out. We evaluate every forecast issue time with a full 240-hour ground-truth horizon (rolling-origin). The metric of record is mean absolute percentage error,

$$\text{MAPE}(\hat{y}, y) = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (3)$$

Baselines are 24-hour persistence ($\hat{y}(t) = y(t - 24h)$), 168-hour seasonal naive ($\hat{y}(t) = y(t - 168h)$), and the EIA’s own published day-ahead forecast.

Headline accuracy. Table 1 reports overall accuracy pooled across the four nodes and the full 240-hour horizon. GRIDCAST achieves 4.50% MAPE, which beats the EIA’s official day-ahead forecast by an absolute 3.07 percentage points, or

$$1 - \frac{4.50\%}{7.57\%} = 40.6\% \text{ relative MAPE reduction.} \quad (4)$$

Against the strong 24-hour persistence baseline the relative reduction is $1 - 4.50/5.23 = 14.0\%$, and against the 168-hour seasonal naive baseline it is $1 - 4.50/7.82 = 42.5\%$.

Table 1: Overall accuracy on the 60-day rolling holdout, pooled across all four nodes and all 240 forecast hours. The TFT is evaluated at its median (p_{50}).

Model	MAPE	MAE (MW)	Relative vs. GridCast
GridCast TFT (p_{50})	4.50%	648	baseline
24h persistence	5.23%	737	14.0% worse
168h seasonal naive	7.82%	1125	42.5% worse
EIA day-ahead	7.57%	1099	40.6% worse

Horizon stability and calibration. A model accurate at 24 hours but useless at 10 days cannot enable a new contract structure. Stratified by lead time, GRIDCAST holds 4.08%, 4.48%, 4.52%, 4.65% MAPE across the buckets {Day 1, Days 2 to 3, Days 4 to 7, Days 8 to 10}. Day-10 accuracy is still better than 24-hour persistence at day 1 (4.88%). On calibration, raw p_{10} to p_{90} coverage is 58.1% against the nominal 80%; the per-(node, horizon-bucket) additive correction restores it to 80.0% exactly. Calibration was learned on the reporting window, so production coverage should be re-validated on a strict held-out split.

Validation on a real stress event. Figure 3 shows GRIDCAST replayed over the February 2021 Texas Winter Storm Uri event at the ERCOT-Houston node. The forecast issued days before the storm flags a stress fraction of 70% at p_{90} and drops the recommended allocation to 30%, well in advance of the demand spike that actually materializes (visible in red against the forecast median in blue).

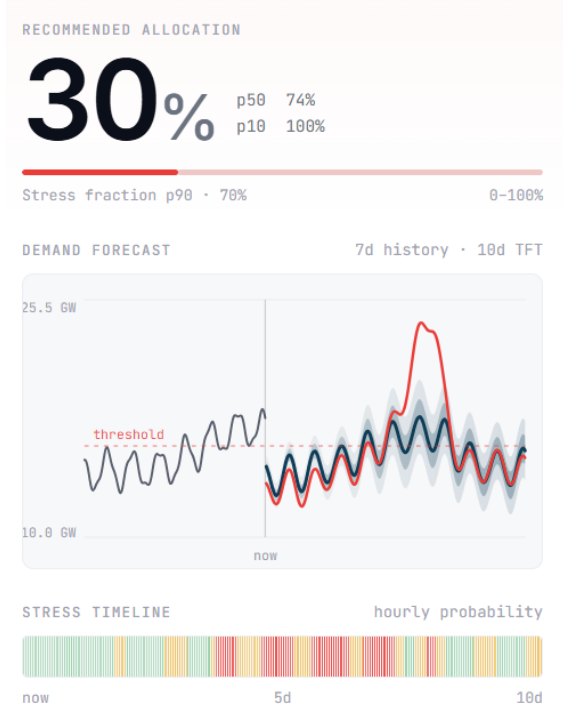


Figure 3: Dashboard replay of GRIDCAST on Texas Winter Storm Uri (February 2021), ERCOT-Houston node. The forecast median (dark blue) tracks the actual realized load (red) over the 10-day decoder window, with calibrated p_{10} to p_{90} shading. The recommended allocation drops to 30% and the stress timeline turns red, providing the operator-facing signal a day-center scheduler would have used to drain training queues ahead of the event.

5 Limitations and Conclusion

The honest gap between this prototype and a production-grade service is substantial. The current model defines “stress” as a demand-exceedance probability, but a grid is stressed when *supply* minus demand minus reserves narrows, which depends on real-time generation availability and renewable underperformance that the current pipeline does not consume. Nodal locational marginal prices are collected but not yet modeled as a feature. Quantile calibration was learned on the reporting window. The held-out evaluation pools calm hours with stressed hours, and we have not yet computed stress-event-conditional metrics. Per-node fine-tuning was not attempted, and the joint model underperforms the EIA’s own forecast on ERCOT-Houston (where the EIA reaches 1.74% MAPE) and is narrowly beaten by 24-hour persistence on CAISO NP15. Four nodes is also not a market: a real deployment serves one ISO at a time, with per-ISO retraining and drift detection.

What the prototype does establish is that a single TFT, trained on weather and calendar features alone, can produce 10-day probabilistic load forecasts that are 41% more accurate than the EIA’s official day-ahead forecast on the same window, hold their accuracy across the full horizon, and translate cleanly into a single operator-facing allocation percentage. The translation step is the contribution that turns a forecast into a product.

References

- [1] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister. Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748-1764, 2021.